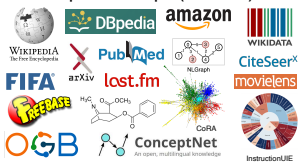


Graph-centric Corpus (about 1.6M)



Input graph from multiple tasks

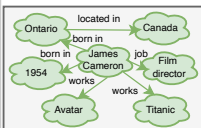
Graph Input Engineering

Instruction & Definition

You are a good graph reasoner / generator, ...

Graph

Passage

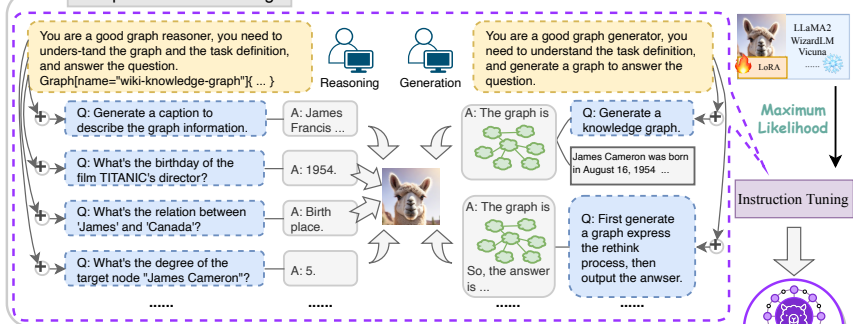


James Francis Cameron CC (born August 16, 1954) is a Canadian film-maker ...

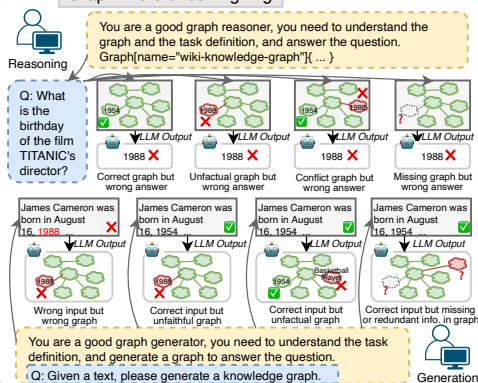
Structured Format Verbalizer

```
Graph [name="wiki-knowledge-graph"]{
  entity_list = [
    "James Cameron", "Ontario",
    "Canada", "1954",
    "Film director",
    "Avatar", "Titanic"
  ],
  triple_list = [
    ("James Cameron" -> "Ontario")[(relation="born in",)],
    ("James Cameron" -> "Film director")[(relation="job",)],
    ("James Cameron" -> "1954")[(relation="born in",)],
    ("James Cameron" -> "Avatar")[(relation="works",)],
    ("James Cameron" -> "Titanic")[(relation="works",)],
    ("Ontario" -> "Canada")[(relation="located in",)]
  ],
}
```

Graph Instruction Tuning



Graph Preference Aligning



Optimization Algorithm (DPO)

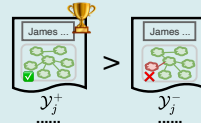
$\mathcal{X}_i$: You are a great reasoner, ...

$\backslash$ Graph[name="..."]{...} \nQ: ...



$\mathcal{X}_j$: You are a great generator, ...

$\backslash$ Graph[name="..."]{...} \nQ: ...



Instruct Graph

Preference Aligning

Maximum Likelihood

